Optogenetics: Concepts and Applications

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Abstract

The high prevalence of brain injuries and mental disorders, the high mortality associated with them, and the inadequacy of current treatment methods for brain injuries all underline the need for new therapeutic technologies. Optogenetics is a combination of genetic engineering and optics, by means of which the activity of cells can be controlled using light; this control takes the form of inhibiting or stimulating a population of cells. The main toolbox of the technology consists of an opsin protein and a light source. This light source can be a laser or a light-emitting diode (LED). Opsins come in various types, and the opsin required for a given application is chosen according to the desired light. The early concepts of the technology date back to the late 1970s, and numerous studies have been carried out on it since. In addition, the clinical applications of optogenetics have gradually increased as the technology has advanced. Most studies of this technology are carried out on the brain and the nervous system. Examples of present-day applications of optogenetics include clinical psychology experiments on animal models using optogenetics, the treatment of neurological and psychiatric diseases such as epilepsy and memory disorders, and studies on the treatment of vision in blind mice. In this review article, we describe the history, concept, and function of optogenetics, as well as some of its clinical applications.

Introduction

Traumatic brain injury (TBI) is one of the leading causes of death and disability. More than 50 million people suffer from TBI every year, and it is estimated that 50% of the world's population will experience a TBI at least once in their lifetime. TBI can affect the lives of people of every age [6]. In general, brain injuries are not limited to TBI alone.

The prevalence of various mental disorders and of suicidal thoughts among those affected underscores the importance of having a treatment strategy [13]. Many studies have reported a fundamental link between severe depression and suicide-related thoughts [12]. A systematic search that used data from 8 major health databases (these data were published between 1 January 2000 and 11 June 2018) links mental disorders and suicide [5]. The search identified 20 articles, of which 13 were unique studies. These studies showed a pooled rate ratio of 13.2 percent for psychotic disorders, 12.3 for mood disorders, 8.1 for personality disorders, 4.4 for substance use disorders, and 4.1 for anxiety disorders; the overall RR for these mental disorders was 7.5, and the population attributable fraction of mental disorders was 21 percent. The results of these findings emphasize the important role of mental disorders in suicide [5].

According to the latest estimate by the World Health Organization in 2019, suicide is one of the leading causes of death worldwide. Each year more people die by suicide than from breast cancer, malaria, war and homicide. In 2019, more than 700,000 people died by suicide. Among young

people aged 15 to 29, suicide was also the fourth leading cause of death, after road injuries, tuberculosis and interpersonal violence [4].

In the treatment of brain disorders and mental disorders by pharmacotherapy, the aim is to increase or decrease the activity of the affected part of the brain; neurological and psychiatric drugs are known as antipsychotic drugs. These drugs have extensive side effects. They are contraindicated in patients with liver damage, cardiovascular disease, Parkinson's disease, people with hypotension or severe hypertension, coma, and severe depressive states, and people with seizure disorders, diabetes, stomach ulcers and chronic respiratory diseases must use these drugs with caution [14].

Drug-based treatments are not always effective, so that for the treatment of bipolar disorders and severe depression, shock therapy is used. ECT (electroconvulsive therapy) is electrical stimulation of the brain while the patient is under anesthesia. It is usually performed by a team of specialized and trained psychiatrists. Like any medical procedure, ECT carries risks, such as loss of short-term memory. Some people have difficulty remembering events from the weeks before treatment; in most cases the memory problems improve after a few months, but some patients may experience problems such as permanent gaps in memory [10].

The treatment and rehabilitation of damaged neural tissue has always been one of the fundamental challenges of the 21st century, owing to the complexity of the central nervous system and its lack of adequate repair capacity after neural tissue injury. The brain, weighing approximately 1300 grams, is made up of billions of neurons and a highly complex network. Because of the complexity of neurons and neural networks, there have always been numerous problems in the precise understanding of neural networks and their function, and of the role of each neural network in human behavior and identity [2].

Optogenetics

Optogenetics is a combination of optics, genetics and bioengineering, with the help of which the activity of cells can be inhibited or stimulated through light-sensitive proteins [2]. Using light, the activity of cells, especially neurons that have been genetically manipulated, can be controlled. Neurons whose genetics have been altered can express a light-sensitive optical channel. Francis Crick was the first to conclude, in order to justify the general theory of the mind, that a method had to be found by which existing neurons of one type could be inactivated without changing the activity of the other neurons [3]. In a separate study in 1973, a paper was published on the properties of a membrane protein in bacteria, in the bacterium Halobacterium. In the following years it became clear that this group of microbial membrane proteins, called opsins, because of the retinal structure within them and the responsiveness of this molecule to light, change the structure of their protein molecule, and this ultimately leads to the passage of some ions through the channel formed in this type of membrane protein [2].

By 2005, various opsins had been discovered in prokaryotes and eukaryotes and their light sensitivity identified, so that in that year, for the first time, a great revolution occurred in neuroscience by introducing a gene expressing microbial opsins into neurons and expressing them in the cells [2]. Optogenetics is a suitable method for observing functional electrophysiological and behavioral changes after optical stimulation. The term optogenetics was coined by Deisseroth in 2006. Until 2007 the technique was used mostly in ex vivo (outside-the-body) settings, but after that, with the invention of optical-neural fiber interfaces, it became possible to study the effect of bacterial opsins on the behavior of healthy, moving animals. From then on, the application of optogenetics

developed rapidly for studying neural circuits, brain diseases and non-neural systems such as stem cells, cardiac tissue and skeletal muscle [3]. From that year onward, the study of the role and function of microbial opsin membrane proteins in cells and neural networks grew remarkably, so that in 2010 optogenetics was named Method of the Year by the journal Nature [2,3]. The ability to study cellular physiology using light-sensitive molecules and genetic coding has profoundly transformed neuroscience [1].

The basic principles of optogenetics in studies and modeling relate to the correct choice of opsin, the method of light delivery, and the specific expression of the gene. Light-sensitive proteins allow researchers to selectively activate and deactivate neurons. Introducing these proteins into cultured cells or into the brain cells of living animals facilitates structural and functional studies of neural networks. The older methods of functional analysis in neurons were based on direct stimulation with very fine electrodes. Although this was an effective method, it was found to be somewhat unreliable because of its limited spatial and temporal precision. With the advent of technology and optogenetic tools, genetically encoded switches gained a higher and more precise ability to turn cells On and Off by means of light control. Functional study of cells within wider network structures will become more feasible with the hope of the emergence and availability of specific opsin proteins [8].

A simple statement of optogenetics is not merely the optical stimulation or inhibition of target cells; rather, optogenetics must deliver the command to the cell to perform or not perform a specific activity. At the basic level, the nervous system can be activated, or can go into a resting state, through a change in the state of an ion pump. For example, activation signals from neuronal carriers cause a flow of positively charged ions into the cell from the outside environment. This activation or non-activation of the cell, resulting from changes in the flow of positively or negatively charged ions, can also be carried out by optogenetics [8].

Manipulating the Membrane Potential

Membrane proteins that change their protein structure following irradiation with a beam of light of a specific wavelength, and thereby make the passage of certain ions easier, are used in optogenetics. Based on the mechanism by which light changes their structure, these proteins fall into two distinct groups:

1. Proteins activated by chemical modification and structural change:
In this group, membrane proteins interact with chemical intermediaries that are sensitive to light, and the occurrence of certain chemical reactions between these intermediaries and the membrane proteins causes structural changes in the membrane proteins and facilitates the transport of ions across the membrane [2].

2. Proteins activated by light:
This group of proteins, owing to the presence of the retinal molecule and its photoisomerization following irradiation at certain specific wavelengths, undergo structural changes and ultimately allow ions to pass. The most important and best-known protein of this group is the opsin. Opsins act as light-responsive ion pumps or as sensory receptors [10]. The tools used in optogenetics for cellular stimulation or inhibition are defined on the basis of light-activated proteins. Depending on the type of change in cell function that researchers need, and also on the speed of action required, various proteins from the microbial opsin family can be used [2].

Microbial Opsins

Microbial opsins absorb the energy of light and use it to pump ions across the cell membrane [10]. These opsins are seven-transmembrane (seven-pass) proteins that are activated by light and are produced widely in archaea, algae and fungi [10,2]. These proteins also give rise to photosynthetic functions.

These opsins fall into two categories:
1. Transporter (pump) rhodopsins
2. Sensory rhodopsins

Transporter rhodopsins include:
1. Chloride pumps (halorhodopsin)
2. Proton pumps (bacteriorhodopsin, archaerhodopsin, proteorhodopsin)



Figure 1: Chlamydomonas reinhardtii

Sensory rhodopsins include:
1. Type I and type II sensory rhodopsins (phototaxis receptors in prokaryotes)
2. Channel rhodopsins (channelrhodopsins) and phototaxis receptors in algae
3. Others, which are responsible for various cellular functions such as movement toward an energy source, moving away from dangerous environments, and controlling the concentration of various ions inside the cell [3].

Introducing Opsins into the Brain

Expression of optogenetic tools in the brain can be achieved in 3 ways:
1. By injection of a viral vector
2. By transgenic modification of animals
3. By in utero electroporation

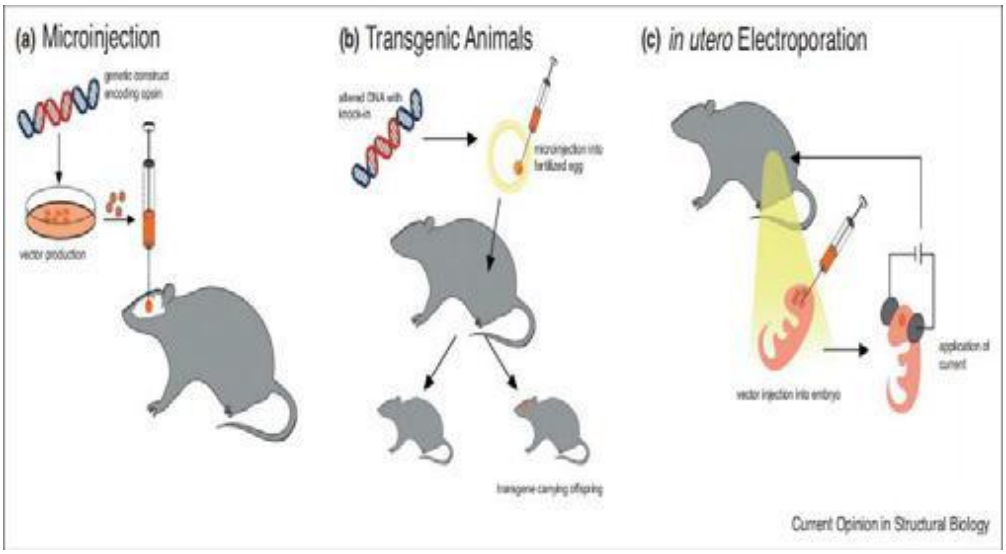


Figure 2: (a) Microinjection; (b) Transgenic animals; (c) In utero electroporation. (Source of the original figure: Current Opinion in Structural Biology)

Transfer of opsin into the mammalian brain:
a. A genetic construct carrying the opsin genes and other elements is placed inside vectors and then injected directly into the brain.
b. Genetically modified material is introduced into the egg cell and injected into females that have been made pregnant by non-natural means, in order to create a transgenic organism.
c. A solution containing opsin DNA is injected into the embryo's brain and is directed, by electroporation, toward a mass of cells [9].

Design of an Optogenetic Study

There are 5 steps for studying the behavior of disease models and normal function:
1. Selecting an appropriate opsin for the experiment
2. Selecting an appropriate targeting strategy, by means of viral delivery methods or vectors that express the opsins in the target cell
3. Selecting optimal light-delivery methods for better penetration of light into the depth of the tissue under study
4. Selecting appropriate optical timing parameters, such as pulse width (duration), frequency and light intensity, according to the light source used
5. Validation, laboratory manipulation, and determination of the performance of the optogenetic tool by means of electrophysiology or immunohistochemistry

In nervous systems, the generation and transmission of neural messages take place through a series of ion channels present in the neuron and through changes in the concentration of ions inside and outside the neuron. In optical stimulation of the nervous system, in the electrophysiology process, light-sensitive proteins are used, in such a way that these protein channels have been created in the neuron by means of genetic techniques. The photocurrent resulting from a light pulse depends on various factors, such as the properties of the opsin used, the wavelength of the light, the intensity and duration of illumination, and even the number of light-sensitive cells at the beginning, middle and end of the process. In general, the important properties of the light sources used in

optogenetics are:
1. The wavelength characteristics required to control the opsin used
2. Temporal control of the light illumination
3. Whether the light must be illuminated in a focused manner or over a broad field [8]

Optogenetic Light Sources

The light sources used in optogenetics are:
1. Laser
2. Diode

1. Laser:
Lasers usually have a suitably narrow spectral width, because by tuning them to the required wavelength a specific tool can be activated. Among opsins, many are sensitive to the wavelengths of blue and yellow light. Red light also has several advantages; for example, red light can penetrate to the depth of tissue [8].

2. Diode:
Another light source used in optogenetics is the light-emitting diode (LED). These diodes have the desired narrow spectral width. The required frequency can also be tuned easily. One of the disadvantages of light-emitting diodes and lasers is the generation of secondary heat when they are close to tissue [7,8].

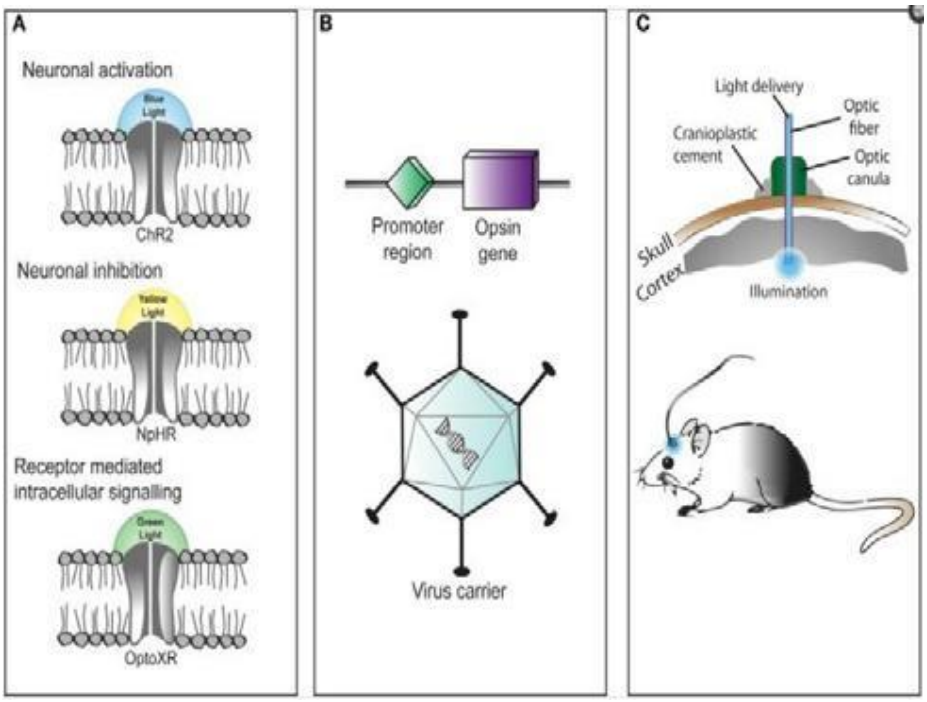


Figure 3: The main components of the optogenetic process:
A. Identification and synthesis of the light-sensitive protein
B. Design of a system for introducing genetic material containing the opsin into the cell, for expression of the protein, such as Cre recombinase
C. Optics (use of a light-emitting device) [8]

Clinical Significance of Optogenetics

With the introduction of various types of opsins, and in view of their speed of action, their return to the baseline state, and their ability to inhibit or trigger action potentials, a wide gateway has opened for the application of optogenetic technology in advancing the treatment of degenerative brain diseases and traumatic lesions [2].

Application of Optogenetics in Clinical Psychology Experiments

In research investigating obsessive-compulsive disorder (OCD) in animal samples, the optogenetic method, by inducing hyperactivity in the OFC-ventromedial striatum pathway, made it possible to evaluate OCD-related behaviors. Excessive stimulation of OFC-Vmc also led to a sharp increase in OCD-related behaviors [13].

Creating False Memories in the Hippocampus

A neurological disorder of human memory can lead to devastating diseases and impair the patient's cognition, to the point that an event is distorted or incorrect information replaces correct information. In order to selectively activate a population of cells, these cells must be isolated and tagged. By forming a rhodopsin channel (ChR2), which is a light-sensitive channel, a pathway for cations can be created. Upon exposure to blue light (470 nm), the neurons in which this channel is expressed are activated.

In this experiment, Tet Tag mice were used, which carry a suitable transgene encoding the c-fos-tetracycline transactivator (tTA). This transgene contains the c-fos promoter, which drives the expression of tTA. tTA can mimic the expression pattern of c-fos and appears transiently in active cells. The tTA protein binds to the tetracycline response element (TRE). This binding is blocked by DOX (doxycycline), which can be administered through the animal's diet. If DOX is removed from the diet, a window for activity-dependent tagging opens, and tTA binds to TRE so that expression of the gene of interest, ChR2-EYFP, is activated only in the activated cells. The transgenic mice were targeted with an AAV vector carrying TRE-ChR2-EYFP, and an optical fiber was implanted directly above the injection site. When the mice become fearful while on a diet without DOX, tTA is expressed like c-fos and enables transcription of ChR2-EYFP; as a result, the cells that generate the sense of fear are tagged, and the mice are then returned to a DOX-containing diet.

Before fear conditioning, habituation procedures for measuring freezing (a behavioral-motor response) in a new context (A) were carried out on animals that were on the DOX diet and in which ChR2-EYFP was therefore not expressed. As expected, the animals showed the lowest amount of freezing behavior in both the light-on and light-off periods. The mice were then taken off DOX, so that a window would open for activity-dependent tagging and for fear conditioning in a different context (B), in order to tag the cells of the fear-memory engram; immediately after this training they were returned to the DOX diet. One day later, the animals were placed again in context A and stimulated with light in order to activate the neurons that had been tagged during fear conditioning in context B, and the behavioral consequences were tested. This tests directly the hypothesis that the c-fos cells of the hippocampus that were activated during training are sufficient for memory recall. Optogenetic reactivation of these cells leads to freezing behavior, which indicates recall of the fear memory.

The hypothesis of whether recall in the light-activated memory is related to a specific context or not was also tested: animals were taken off the DOX diet in context A, and the DG (dentate gyrus) cells were tagged with ChR2-EYFP. The animals were then placed on DOX again so that no further tagging would occur. The next day they were frightened in context B; these animals thus had both a non-fear memory engram labeled with ChR2 for the neutral context A and an unlabeled fear-memory engram for context B. These observations argue that memory engram cell populations are specific to DG tissue. Reactivation of the DG cells that were active during fear-conditioning training is sufficient to confer recall of the fear memory.

This system can target specific regions of the brain, specific cell types, and also the cell population involved in a specific memory. The remarkable ability of optogenetics focuses on behavior-related cell ensembles and makes circuit mapping of memory engrams across the brain possible. Hippocampal engram cell populations, after being activated by optical stimulation, can recall related memories [16].

Optogenetic Studies of Networks in Epilepsy

In the mid-twentieth century, researchers used pharmacological and electrical tools to study epilepsy networks. This research led to a remarkable growth in the treatment of epilepsy. However, approximately 30 percent of patients with epilepsy do not respond to drug treatment. The development of optogenetics, owing to its unprecedented precision in neuroscience, has made it possible to target specific cell types and, by turning them on or off with precise temporal control, to reveal the details of the seizure circuit. Of course, the use of optogenetics in epilepsy research is still in its early stages [18].

Optogenetics: Restoring Vision in Blind Mice

Expression of light-sensitive microbial opsins is a promising approach for restoring vision in degenerative retinal diseases without the need for invasive surgery. The main idea of optogenetics here is to convert the light-sensitive neurons of the retina into artificial photoreceptors. One of the existing strategies is to bypass the photoreceptors and target the bipolar cells. When the bipolar cells degenerate, artificial photoreceptors are made from ganglion cells. Through expression of ChR2 in retinal ganglion cells, vision can be restored after the photoreceptors have completely degenerated. Non-selective expression in retinal ganglion cells does not allow image processing by means of the retinal neurons. Targeting retinal bipolar cells with microbial opsins can produce ganglion cell responses that are closer to the pattern of normal activity. In an important step, electroporation was used to express ChR2, under the control of the ON bipolar cell promoter, in the ON bipolar cells of blind mice (rd1). ChR2-mediated activation was induced in the bipolar cells. Sudden responses in retinal ganglion cells led to the restoration of visually evoked potentials in the cortex and to control of visual behaviors [10].

In one study on the optogenetic treatment of the visual abnormality retinitis pigmentosa, which destroys the eye's photoreceptors, the lack of photoreceptors in the eye was compensated for by inducing light sensitivity in retinal ganglion cells [15].

Conclusion

Optogenetic technology has expanded into various fields. The progress of this technology in understanding and studying the role of neurons, neural circuits and the connections between them

has made it possible that, in the near future, we will witness wider use of optogenetics as a method for treating neurological diseases, brain disorders and traumatic injuries.

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(Translator's note: the printed magazine lists only references [1]–[7]; the in-text citations [8]–[18] have no corresponding entries in its reference list.)